

A Geometric Explanation of The Frisch-Waugh-Lovell Theorem

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We provide a geometric explanation of the well-known Frisch-Waugh-Lovell (FWL) theorem without delving into the setup details. Suppose we have the following linear regression

$$Y = X_1\hat{\beta}_1 + X_2\hat{\beta}_2 + \hat{\epsilon}, \quad (1)$$

with $\hat{\beta}_1$, $\hat{\beta}_2$ and $\hat{\epsilon}$ being the OLS estimates, and the following regressions

$$X_1 = X_2\hat{\gamma}_1 + \hat{\eta}_1, \quad (2a)$$

$$Y = X_2\hat{\gamma}_2 + \hat{\eta}_2, \quad (2b)$$

with $\hat{\gamma}_1$, $\hat{\gamma}_2$, $\hat{\eta}_1$ and $\hat{\eta}_2$ being the OLS estimates.

The FWL theorem states that if we regress $\hat{\eta}_2$ on $\hat{\eta}_1$, the OLS estimate of the coefficient is equal to $\hat{\beta}_1$, and the residual of the regression is equal to $\hat{\epsilon}$, i.e., with OLS estimates we have

$$\hat{\eta}_2 = \hat{\eta}_1\hat{\beta}_1 + \hat{\epsilon}. \quad (3)$$

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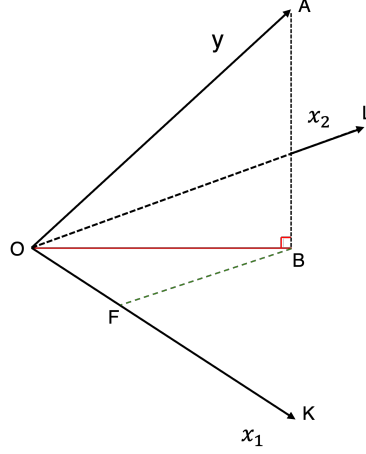


Figure 1: Geometry of OLS: $\hat{\epsilon}$ is represented by the segment AB perpendicular to the X_1 - X_2 plane; and $X_1\hat{\beta}_1$ is represented by the segment OF with BF parallel to X_2 . Thus, $\frac{|OF|}{|OK|} = \hat{\beta}_1$.

We first recall the geometric interpretation of the OLS estimates for Equation (1). In Figure 1, the line segment OA represents Y , the segment OK represents X_1 , and the line segment OL represents X_2 . If we draw a segment AB perpendicular to the X_1 - X_2 plane, then $X_1\hat{\beta}_1 + X_2\hat{\beta}_2$ is represented by the segment OB, and $\hat{\epsilon}$ is represented by the segment AB. If we draw a segment BF parallel to X_2 , then OF represents $X_1\hat{\beta}_1$. Thus, $\frac{|OF|}{|OK|} = \hat{\beta}_1$.

Let's consider the geometric interpretation of Equations (2a) and (2b). In Figure 2, we do the following:

- Draw a line segment AC perpendicular to the X_2 (line segment OL). Then OC is the projection of Y onto X_2 , i.e., $X_2\hat{\gamma}_2$, and segment AC represents $\hat{\eta}_2$.
- Note that in this case, X_2 is perpendicular to the plane ACB.
- Draw a line segment KD perpendicular to the line segment X_2 (segment OL). Using similar reasoning as above, KD is the projection of X_1 onto X_2 , i.e., $X_1\hat{\gamma}_1$, and segment KD represents $\hat{\eta}_1$.

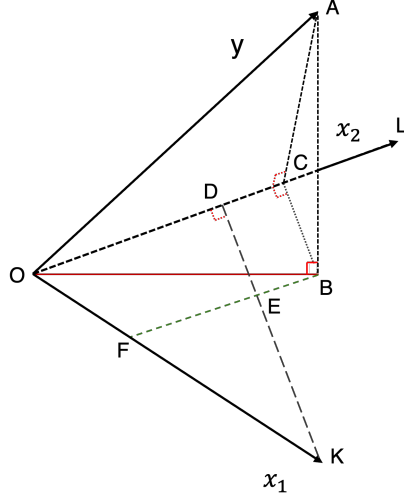


Figure 2: Interpretation for FWL theorem.

Now consider Equation (3) in Figure 2:

- Since segment KD is parallel to segment BC, and BF is parallel to segment OL, the length of DE is equal to the length of BC: $|DE| = |BC|$.
- Accordingly, segment BC is the projection of $\hat{\eta}_2$ onto $\hat{\eta}_1$, and segment AB represents the OLS residual when regressing $\hat{\eta}_2$ on $\hat{\eta}_1$.
- Because segment BF is parallel to segment OL, we have

$$\frac{|BC|}{|OK|} = \frac{|DE|}{|DK|} = \frac{|OF|}{|OK|} = \hat{\beta}_1.$$

- Thus, $|BC| = |DE| = \hat{\eta}_1 \hat{\beta}_1$.

As segment AB represents $\hat{\epsilon}$, we have validated the Equation (3).